

JOONWON LEE

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SUMMARY

Statistics Ph.D. candidate and quantitative researcher specializing in hypothesis-driven empirical research, scalable likelihood-based modeling, econometrics, machine learning, simulation, and statistical diagnostics. Builds reproducible Python/PyTorch pipelines for large, complex datasets and evaluates when apparent signals reflect persistent structure versus noise or model misspecification. Brings financial-services experience and a strong interest in systematic equity research.

EDUCATION

Ph.D. in Statistics, Rutgers University, Piscataway, NJ

Expected May 2027 | GPA: 3.8/4.0

M.S. in Statistics, University of Minnesota, Minneapolis, MN

Sep 2021 | GPA: 3.7/4.0

TECHNICAL SKILLS

Programming and Research Systems: Python (NumPy, Pandas, SciPy, PyTorch, LightGBM, scikit-learn), SQL, C++/pybind11 integration, Linux, Git, AWS EC2, HPC/SLURM; CPU/GPU computing and restartable analytical pipelines

Quantitative Research: Hypothesis testing, econometrics, regression, GLM/GBM, machine learning, stochastic processes, MLE/GLS, Monte Carlo simulation, numerical optimization, spectral analysis, feature attribution, model selection, validation, sensitivity analysis, and performance evaluation

EXPERIENCE

JPMorgan Chase - Summer Quantitative Analytics Associate, Model Risk Governance and Review

Summer 2026

- Developed and validated a Sequential Probability Ratio Test for binary risk indicators, converting a monitoring question into explicit hypotheses, likelihood-ratio boundaries, and controlled sequential decisions.
- Combined Brownian-motion approximation, dynamic-programming probability propagation, and simulation to evaluate boundary-crossing probabilities, stopping behavior, Type I/II error, and early-decision tradeoffs.

Travelers - Data Science Leadership Program, Business Insurance Pricing Team

Summer 2025

- Developed an end-to-end LightGBM pipeline across 2.46M+ records, including data preparation, feature processing, GLM benchmarking, model selection, validation, and performance diagnostics on AWS EC2.
- Evaluated predictive performance and feature-attribution patterns and presented pricing and risk-segmentation recommendations to business stakeholders.

QUANTITATIVE RESEARCH AND MARKETS

IMC Prosperity Algorithmic Trading Competition - Top 2.39% of Participants

Apr 2025

- Built and evaluated a market-making strategy using market-based fair-value estimation, inventory-aware quoting, rolling-window exposure controls, and benchmark-driven performance analysis.

Ph.D. Dissertation Research - Scalable Statistical Research and Diagnostics

Sep 2024 - Present

- Scalable Statistical Modeling:** Developed an advection-aware Vecchia likelihood approximation for nonseparable Gaussian processes, replacing dense covariance operations with ordered local conditioning to fit up to 145,008 observations per day.
- Hypothesis Evaluation and Diagnostics:** Designed simulation studies and scale- and frequency-resolved tools to distinguish persistent dependence from low-frequency misspecification, high-frequency noise, and data artifacts.
- Research Pipeline:** Built reusable Python/PyTorch workflows for data-quality filtering, missingness tracking, time-dependent alignment, estimation, simulation, cached outputs, and restartable CPU/GPU and HPC execution; integrated compiled C++ routines through pybind11.

Econometrics Project - Earned Income Tax Credit and Household Labor Supply

Graduate Research

- Estimated the effect of the Earned Income Tax Credit on household labor supply using fixed-effects regression and propensity-score matching with observed household covariates.

ADDITIONAL

Coursework: Econometrics, Machine Learning, Probability Theory, Stochastic Processes, Statistical Computing, Advanced Theory of Statistics I-II, Data Structures and Algorithms, Linear Algebra

Languages: Korean (Native); English (Fluent); Chinese (Basic)